MB504T Final Term 2024

Advantage of RAPD:

1- No need for specific primers or DNA probes
2- Quick and efficient process
3- Small DNA amount required
4- High fragment yield 5-. Low costs

Name the enzyme used in joining DNA fragments:

The enzyme used in joining DNA fragments is called DNA ligase. DNA ligase plays a crucial role in the process of DNA replication and repair by catalyzing the formation of phosphodiester bonds b/w adjacent DNA fragments.

Techniques to get unknown Polymorphism:

* Direct Sequencing * Microarray * Cleavage/Ligation
* Electrophoretic mobility assays

Transfer Method of DNA to membrane: (Southern Blotting)

* Upward Capillary transfer * Downward Capillary transfer
* Simultaneous transfer to two membranes
* Electrophoretic transfer * Vacuum transfer.

Limitations of Western Blotting:

Delicate Process, incorrect labeling, insufficient transfer, trained technicians needed, semi-quantitative estimation.

Limitation of Northern Blotting:

time-consuming, requires large RNA amount, less sensitive, limited detection, experimental errors, subjective interpretation.

Adjvants and it's example: or reduce non-specific binding

Adjuvants are substances added to vaccines to enhance the immune response. e.g; aluminium salts. ELISA W.B

Joining DNA Molecule by alkaline Phosphatase:

Alkaline Phosphatase is an enzyme that removes phosphate groups from molecules like DNA. Treating linearized plasmid vector DNA with 5'-terminal phosphate groups, preventing recircularization and plasmid dimer formation.

Screening Techniques of SNPs:
Two different screening strategies:
Many SNPs in a few individuals
A few SNPs is many individuals
Methods of Joining DNA Fragments: (3 methods)
1- Joining covalently annealed cohesive ends using DNA ligase.
2- Joining blunt-ended fragments using DNA ligase from phage T4 infected E.coli.
3- Terminal deoxynucleotidyltransferase can synthesize homopolymeric 3' single stranded tails at the end of fragments.

RFLP:
RFLP analyzes DNA by measuring fragment length. DNA is cut by a restriction enzyme and separated on a gel (gel electrophoresis). Fragments are transferred (via Southern blot) to a membrane and hybridized with a labeled probe. Variations in fragment length are used for genetic analysis.

VNTR & its types:
DNA profiling uses repetitive sequences called VNTRs or STRs to determine relatedness. VNTRs are short sequences that repeat in the genome and show variation in length b/w individuals. They are used in DNA fingerprinting for genetic analysis.
There are 2 types of VNTRs;
Microsatellites (SSRs or STRs)
Minisatellites (VNTRs)
They are valuable in genetics, forensics, and biology research.

Haplotype correlation to phenotype:
Combining adjacent SNPs into multilocus haplotypes captures regional LD information for powerful association | with phenotypic frequencies.

Linkage Disequilibrium

	Northern Blotting	Southern Blotting	Western Blotting
Types of Molecule detected	detects specific RNA seq from RNA sample.	detects specific DNA seq from DNA sample	dectects protein from a protein sample.
Types of Gel	Agarose & Formaldehyde	Agarose gel	Polyacrylam gel
Blotting Method	Capillary transfer	Capillary transfer	Electric transfer.
Probes Used	cDNA or RNA probes labeled non-radioactively or radioactively	DNA probes labeled radioactively or non-radioactively	Primary antibodies are used as probes.
Detection System	done using an autoradiograph or detection of light or color change.	done using an autoradiograph or detection of light or color chang	done using the detection of light or color change.
Applications	Used to study gene expression by detecting RNA molecule. or analyzing	in genetic testing, DNA fingerprinting and studying gene mutation or variations.	Used to study gene expression, protein-protein interaction & protein modifications.

Steps of Insit PCR:
Fixation, Permeabilization, Denaturation, Annealing, Extension, Amplification, Detection.

Techniques to get known Polymorphism:

* Hybridization techniques: Microarrays, Real time PCR.
* Enzyme based techniques: Nucleotide extension, Cleavage, ligation, Reaction production dection and display.

Radioactive & Non radioactive Probe characteristics

OR. Southern Blotting Probes:

* Radioactive Probe - P32
Sensitive Relatively Cheap Hazardous
Radioactive waste disposal regulations should be followed.
* Non-Radioactive Probe - Biotin
Sensitive Relatively Expensive.

Characteristics of SNPs:

1- In digestion diagnosis 3- In drug discovery & Development
2- In finding predisposition to disease 4- In Drug Responses
& Investigation of Migration Patterns.

Steps of fingerprinting

1- Isolation the DNA 2- Cutting, Sizing, Fixing
3- Transfer of DNA to nylon 4- Probing 5- DNA fingerprinting.

Which enzyme is used in mini primer PCR:

Mini Primer PCR uses thermostable Polymerase (S-Tbr) that can extend from short primers as short as 9 or 10 nucleotides

DNA modifying Enzyme:

Nucleases DNA Polymerase Revelse Transcriptase DNA Ligase.

Electron Microscope:

Electron microscope use electrons instead of light to magnify tiny objects with high resolution.

Uses of Microscope or Microscopy:

Many uses in biology, medicine, material science and nanotechnology.

Isoelectric and Gradient Gel:

Isoelectric focusing separates proteins based on their isoelectric points using a gel with a pH gradient.
Gradient gel creates a concentration gradient to separate molecules.
Both are useful techniques in protein analysis.

Uses of Electron Microscope:

Electron Microscope are used to study cells, materials, and nanoscale structures with high resolution.

Immunoflourescent Microscope:
helps visualize specific protein or molecules within cells using flourescently labeled antibodies
Types of Electron Microscopes:
Transmission electron microscopy (TEM) and Scanning electron microscopes (SEM) are two types of electron microscopes. TEM creates images of internal structure while SEM focuses on surface details. They provide high resolution images.
Ion Exchange Chromatography:
Ion exchange chromatography separates and purifies charged molecules based on their interaction with charged stationary phases.
Principle of Southern blot & how it cleaved the DNA:
Southern blot is a technique used to detect specific DNA fragments in a sample. In Southern blotting, DNA is cut into fragments using restriction enzymes for analysis.
Double Sandwich antibody ELISA
Double sandwich antibody ELISA is a method to detect specific protein using two antibodies
ELISA: Enzyme linked immunosorbent assay.
is widely used technique in molecular biology and immunology. It is used to detect and quantify specific proteins or molecules in a sample. ELISA involves the use of antibodies that bind to the target molecules.
Allelic Specific PCR:
Allelic Specific PCR detects specific (PCR) gene alleles using specially designed primers for selective amplification.
Primers:
Short DNA sequence that start DNA replication or amplification.
Misprimings: May occur due to hybridization of primers to each other
Primer Dimers: May reduce amplification
Misprime may occur due to non-specific hybridization of primers.

① dNTPs: *

* Deoxynucleoside triphosphates (dNTPs) are the building blocks from which the DNA polymerase synthesizes a new DNA strand during successive cycles of PCR amplification.
* dNTPs consist of four basic nucleotides - dATP, dGTP, dCTP, and dTTP. The concentration of each dNTP in the PCR is usually 200uM.

② Nested and Multiplex PCR: *

Nested PCR: Use to increase the specificity of DNA amplification.

* Uses two pairs of primers, with the second pair (Nested PCR Primers) binding inside the first PCR product.
* Producing a shorter and more specific second PCR product
* Very low probability of nonspecific amplification.
* Increasing the specificity of DNA amplification

Multiplex PCR: is a variant of PCR, allows the simultaneous amplification of multiple targets

* Using more than one pair of primers in a single reaction.
* Amplicons of different sizes are produced.

③ Amplified Fragment length Polymorphism (AFLP): *

* AFLP involves genomic DNA digestion with restriction enzymes (tetracutter (Mse1), Hexacutter (EcoR1)).
* Ligation of linkers to restriction fragments, Pre-selective PCR using specific primers, electrophoretic separation, and visualization of band patterns.

*AFLP is a highly sensitive PCR-based method for detecting DNA polymorphism
*Can also be used for genotyping individuals for a large number of loci.

④ Hot Start PCR:
It is a technique performed manually by heating the reaction components to the DNA melting temperature (e.g 95°C) before adding the polymerase.

⑤ Mini-Primer PCR:
*Mini-Primer PCR uses a thermostable polymerase (S-Tbr) capable of extending from very short primers (9 or 10 nucleotides)
*This method allows for PCR targeting to smaller primer binding regions.
*Commonly used for amplifying conserved DNA sequences. like 16S (or eukaryotic 18S) rRNA gene.

⑥ Limitations of PCR:
*Contamination Risk *Primer Complexities * Primer-binding site Complexities
*Amplifies rare species *Detection (~~Risk~~) Method.

⑦ Methylation PCR: (MSP) (specific)
*Used to identify patterns of DNA methylation at cytosine guanine island (CpG islands) in genomic DNA.
*CpG islands, are concerned in regulation of gene expression in mammalian cells.
*Target DNA is first treated with sodium bisulfite, which converts unmethylated cytosine bases to uracil, which is complementary to adenosine in PCR primers.
*Two PCR reactions are performed: one targeting methylated cytosines and the other targeting unmethylated cytosines.
*MSP used in quantitative PCR provides information about methylation state of specific CpG islands.

⑨ Conventional PCR (Real time PCR or q-PCR) $ Ingredients *

* Conventional PCR also known as real-time PCR or q-PCR.
* is a technique used to amplify and detect specific DNA sequences
* Ingredients: DNA template, primers (forward and reverse), nucleotides (A, T, C, G), DNA polymerase, and a buffer solution.
* These ingredients work together to amplify the DNA of interest.

⑩ Colony PCR:

* Amplifies DNA directly from bacterial colonies.
* Quick screening method to confirm the presence of a specific DNA seque
* Uses a single colony as the template for PCR.

⑪ In Situ PCR: limited to detection of non-genomic material such as RNA.

* Amplifies DNA within cells or tissues.
* Combines DNA amplification and visualization of the target sequence in the cellular context.
* Used for studying gene expression and localization.

CamScanner

(14) Stages of PCR: *

Denaturation: Double stranded DNA melts into single stranded DNA.
* DNA sample is heated to high temperature 93°C to 98°C, typically 95°C for 20-40 seconds.

Annealing: Primers bind to complementary DNA sequences.
*Step is usually done around 40-65°C for 20-40 seconds.
Annealing of primers to single-stranded DNA template.

Extension: DNA polymerase adds nucleotides, synthesizing new DNA strands.
*DNA polymerase binds to annealed primers & extend DNA strands at the 3' end of the chain.
*Typically around 70°C to 80°C ⇒ Commonly 72°C for 30 sec to 2mins

These three stages are repeated multiple times in a PCR machine, resulting in exponential amplification of target DNA sequence.

(15) Types of R-M (Restriction Modification System): *

Type I R-M System: were the first to be characterized from E.coli K12.
*The active site consists of two restriction subunit, two modification subunit and one recognition subunit.

Type II R-M System: Well known and widely used *separate restriction and modification enzymes *specific recognition and cleaves it at defined site.

Type III System: Separated activities
*Variable cleavage sites
*Require DNA methyltransferase protein.
∴ Type III enzymes have symmetrical recognition sequences but are little value.

Type IIs:
Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness.

Type I system are of little value of gene manipulation

(7) Clinical Microbiology Applications: *

1 Identification of bacteria: *Slow growing bacteria
*Bacterial antibiotic resistance genes.

2 Virus: almost all viruses identification and viral load.

3 Fungus: Aspergillus spp *Candida spp *Plasmodium spp
*Trypanosoma spp *Babesia spp

(8) Forensic Applications of PCR:

*DNA fingerprinting *Drug discovery *DNA profiling
*DNA typing *Gene therapy *Mutation detection
*Classification of organism *Genotyping *Detection of pathogens
*Prenatal diagnosis of genetic disorders. *Detection of mutation
*Mutation Screening *DNA abnormalities *Mitochondrial DNA analy

(9) Reagents use in Standard PCR: *

Generally, a complete PCR requires five basic reagents, includi

1- DNA Template 2- Forward and Reverse Primers
3- DNA Polymerase 4- Deoxy nucleotide Triphosphates (dNTPs)
5- Reaction Buffer

(26) WGA (Whole Genome Amplification) PCR:

* is a technique used to amplify the entire genome of an organism.
* It involves amplifying small amounts of DNA into larger quantities allowing for downstream analysis.
* Commonly used when the starting material is limited such as in forensic or clinical samples

(27) Advantages of Asymmetric PCR: 2 (Advantages)

1 Generates a lot of single-stranded DNA
2 Selectively amplifies one strand.
It great for sequencing and studying specific regions.

(28) Chemicals used in Conventional PCR:

Microfuge tube, Thermal cycler, DNA template, Primers, Buffers, $MgCl_2$, Distilled H_2O, Deoxynucleotide triphosphates, DNA polymerase.

(29) Methodolog of QRT-PCR:

QRT-PCR methods use flourescent dyes, such as Sybr Green, or fluorophore-containing DNA probes, such as TaqMan, to measure the amount of amplified product in real time.

(30) Reverse-Transcriptase PCR:

RT-PCR also known as reverse-transcriptase PCR, is a variant of PCR that typically measure RNA expression levels. and also reverse transcribes the RNA into DNA.

(31) RAPD (Randomly amplified Polymorphism DNA): *

is a PCR-based technique which uses arbitrary primers which bind to nonspecific sites on the DNA and amplify the DNA. These amplified fragments then migrated on agarose gel and diff inband pattern is observed.